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Prior-Knowledge-Guided Dual-Stream Fusion Network for Few-Shot Medical Image Classification

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ABSTRACT

Few-shot medical image classification focuses on learning effective feature representations from limited labeled data, however, this task remains challenging due to domain shifts and high inter-class similarity commonly observed in medical imaging. Recent vision–language models, such as contrastive language–image pretraining (CLIP), utilize large-scale semantic knowledge and have demonstrated significant potential. However, when applied to medical data, CLIP-based methods tend to emphasize global semantic alignment, thereby failing to adequately capture fine-grained pathological details that are crucial for reliable diagnosis under limited supervision. To address this limitation, we propose a prior knowledge-guided dual-stream fusion network (PKDNet) for cross-domain few-shot medical image classification. PKDNet retains CLIP's powerful vision–language semantic priors to provide robust global context, while introducing an auxiliary convolutional branch to explicitly capture local structural and textural information. The two streams are fused through an uncertainty-aware mechanism that adaptively integrates complementary semantic and structural representations. Extensive experiments on the MedFMC benchmark show that PKDNet consistently outperforms existing approaches, achieving improvements of 10.82%, 55.74%, and 0.5% under the five-shot setting across three datasets. These results highlight the importance of explicitly combining semantic and structural information when adapting vision–language models to data-scarce medical imaging scenarios.

1 | Introduction

Few-shot learning (FSL) is a fast-growing branch of machine learning that focuses on categorizing previously limited sets of labeled samples with few labels. Recently, FSL has focused on computer vision, natural language processing, and robotics. However, with fewer labeled examples, models tend to overfit and fail to learn truly generalizable features. This issue is especially pronounced in medical imaging, where subtle variations across disease stages make it difficult to establish robust decision boundaries from limited data.

Existing FSL approaches can be broadly categorized into four classes based on their different methods for alleviating data scarcity: transfer-based methods, meta-learning methods, data augmentation methods, and multi-modal methods [1]. Transfer-based approaches adapt models trained on large-scale natural image datasets to downstream tasks via fine-tuning [2–4]. The problem transfer-based approaches face is the large domain gap between medical and natural images; as a result, they often lead to negative transfer and limited adaptability in clinical contexts.

Meta-learning methods were developed to mitigate the limitation of inter-domain transferability due to significant differences

Abbreviations: ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor.

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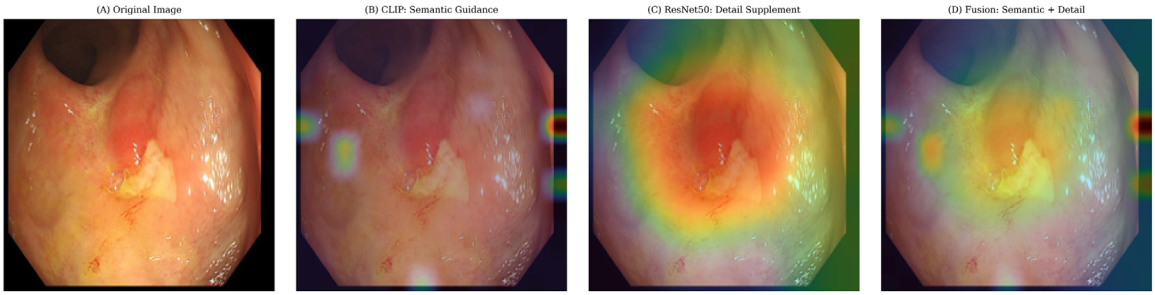


FIGURE 1 | Qualitative visualization of semantic, structural, and fused representations for pathological image analysis. (A) Original pathological image after inverse normalization for visualization. (B) Semantic attention map generated by the vision–language model CLIP, highlighting lesion-level regions guided by cross-modal supervision. (C) Structural response map extracted from the trained ResNet50, emphasizing local textures and boundary details. (D) Fused representation combining CLIP-derived semantic guidance and ResNet-based structural features, yielding more comprehensive and discriminative lesion characterization.

between domains. Their goal is to learn transferable representations from many tasks so that you can quickly apply them to previously unseen classes based on very few labelled images [5, 6]. Although meta-learning techniques are conceptually appealing, they are expensive computationally to train, sensitive to how tasks are defined and have limited ability to be applied in different clinical settings.

Data augmentation techniques are designed to address the lack of available training images by synthesising new samples or applying different transformations to them [7, 8]. These techniques help to reduce the risk associated with insufficient data, but still create the risk of generating inappropriate and noisy samples. This poses an issue in the field of medical imaging, where it is especially critical to identify variations in anatomy and diagnostic accuracy at a very detailed level. These limitations have motivated increasing interest in incorporating auxiliary semantic knowledge into medical image analysis. Recent studies suggest that language can serve as an effective complementary source of supervision, enhancing both recognition and reasoning capabilities under limited labeled data [9].

Recent advances in medical image analysis have shown a clear shift from conventional multimodal fusion strategies toward general-purpose vision–language foundation models, which provide transferable semantic priors for a wide range of downstream tasks under limited supervision [10]. This trend is particularly relevant to few-shot medical image classification, where incorporating textual knowledge can alleviate the lack of annotated data and improve representation generalization.

Against this background, multimodal frameworks that integrate complementary information from vision and language have become an increasingly important research direction. In particular, vision–language models (VLMs) such as contrastive language–image pretraining (CLIP) inject high-level semantic priors, such as disease descriptions or anatomical concepts, into visual representation learning, thereby enriching feature expressiveness with domain-relevant knowledge [11, 12]. Recent work has further demonstrated the effectiveness of this paradigm in few-shot medical image classification. For example, PM^2 introduces a prompting-based multimodal framework in which textual prompts provide complementary semantic guidance for image representation learning, leading to improved performance under limited supervision [13].

Despite these advantages, existing VLM-based methods still exhibit important limitations in medical imaging. CLIP builds strong semantic awareness of words and sentences and can effectively localize lesion-related regions through cross-modal supervision (see Figure 1). However, the spatial distributions of attention in VLMs are often smooth and relatively insensitive to subtle structural variations, even though such variations are biologically significant for lesion morphology characterization. In contrast, convolutional neural networks, such as ResNet50, are better at capturing local textures and boundary information, but they often lack explicit semantic constraints, which may lead to spatially diffuse activations and susceptibility to non-pathological structures.

We can conclude that these observations indicate that semantic- and structure-driven representations are complementary in many ways; however, we must be careful not to use either approach exclusively to characterize pathology in its entirety, as neither is sufficient to provide comprehensive and reliable pathological understanding.

In this work, the term “cross-domain” refers to few-shot evaluation across heterogeneous medical imaging domains, rather than strict source-to-target cross-dataset adaptation. Specifically, the considered datasets cover chest X-ray imaging, endoscopic imaging, and histopathological tissue imaging, which differ substantially in imaging modality, acquisition device and protocol, anatomical region, visual distribution, label space, and task formulation. Therefore, our goal is to evaluate whether the proposed semantic–structural fusion strategy remains effective under substantial domain heterogeneity when only limited labeled samples are available.

To address these limitations, this paper proposes a prior knowledge-guided dual-stream fusion network (PKDNet) that effectively combines the global semantic understanding of CLIP [14] with the domain-specific feature extraction capabilities of a customized ResNet architecture. In this dual-stream framework, image features are processed in parallel through two complementary pathways: one that leverages CLIP’s vision-language pre-training to capture high-level semantic represen-

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tations aligned with clinical text, and another that employs a structurally enhanced ResNet, specifically adapted for the fine-grained requirements of medical image analysis. The enhanced ResNet is designed to preserve detailed spatial information and improve discriminative precision through three architectural refinements. First, the network introduces a deep convolutional stem with multiple small-kernel layers and average pooling, which enhances low-level texture representation. Second, anti-aliasing strided convolutions are employed to reduce feature distortion and maintain spatial consistency across downsampling operations. Third, instead of conventional global average pooling, a self-attention mechanism with query, key, and value projections is applied to enable the model to dynamically attend to diagnostically salient regions within the image. These modifications enhance the network's ability to extract clinically relevant features under few-shot conditions. The final prediction logits are generated by fusing the global semantic embeddings from CLIP and the localized visual features from the enhanced ResNet branch. This design integrates CLIP's strong zero-shot generalization with the domain adaptability and fine-grained sensitivity of convolutional representations, effectively mitigating domain shift and reducing dependency on large-scale labeled datasets. Moreover, the attention-guided localization of pathological regions mirrors radiologists' diagnostic reasoning, enabling more accurate and context-aware classification in the medical imaging domain.

The contributions of this paper can be summarized as follows.

- This paper proposes a PKDNet that integrates CLIP's global semantic understanding with a structurally enhanced ResNet for fine-grained medical image analysis, effectively addressing domain shift and improving few-shot classification performance.
- The enhanced ResNet branch incorporates a deep convolutional stem with small-kernel layers, anti-aliasing strided convolutions, and a self-attention-based pooling mechanism, enabling precise localization of pathological regions and mimicking radiologists' diagnostic reasoning process.
- Extensive experiments validate the framework's superiority in cross-domain FSL, particularly in medical image analysis, achieving significant improvements in accuracy and robustness.

2 | Related Work

2.1 | Few-Shot Learning

FSL [15] methods can be divided into four categories, including transfer-based methods, meta-based methods, data augmentation-based methods and multi-modal-based methods.

2.1.1 | Transfer-Based Methods

The goal of transfer learning is to leverage knowledge from one or more source tasks when developing a new target task with little or no high-quality training data [16]. Traditionally, an early example of this approach is "TrAdaBoost," created by Dai et al. [17], which leverages a small amount of labeled target data to reweight source instances and reduce negative transfer when the source and target distributions differ. Although these methods worked well for simple shifts in domain, they were ineffective for high-dimensional visual data because they relied on shallow representations of feature instances, which prevented them from accommodating the complexities of relationships between cross-domain data sources [18]. Therefore, there was a need to adopt deep learning approaches to develop an effective transfer learning model that could better leverage the network's ability to learn features across multiple layers. The introduction of vision transformers (ViT) has been a major advancement in this area because of the pre-training process used for training over large databases like ImageNet, which has shown that it is possible to combine self-attention with linear detection in order to achieve superior results in transfer learning when using a few images to train [19]. Unlike previous methods that were instance-based and required an alignment of the source and target distributions, the use of ViTs provides a method to learn representations that are not task-specific through the use of a model that is aware of all instances in the dataset and has been trained using self-attention, thus allowing for greater flexibility in the transfer of knowledge between multiple datasets. ViTs, however, remain reliant on pre-training with supervised data to perform optimally in few-shot learning scenarios, limiting their ability to scale in settings with scarce labels. To address this limitation, several self-supervised learning methods have been developed to leverage contrastive learning techniques and achieve the same level of knowledge and representation transfer capability as supervised training without annotated data [20]. These self-supervised methods effectively combine the strengths of ViTs trained with supervised learning and provide a means to bridge the gap between these models and real-world applications by using unsupervised pre-training to learn transferable representations at scale, while using the techniques and benefits of transformer architectures.

2.1.2 | Meta-Based Methods

Meta-learning, often described as 'learning to learn,' is a machine learning methodology that enables a system to improve its own learning process [21]. It achieves this by training on a variety of tasks, which allows a base learner to acquire generalizable knowledge or strategies for rapid adaptation to new tasks. Through systematic refinement of the learning procedure, including how models are initialized and adapted, meta-learning enhances the rate of knowledge transfer and sample efficiency in few-shot learning scenarios [6]. The field aims to build adaptive systems that can quickly learn new tasks with minimal data. It draws inspiration from both automatic machine learning (e.g., efficient hyperparameter optimization) and cognitive science (e.g., theories of human skill acquisition). By leveraging insights from these areas, meta-learning creates systems that can adjust their learning strategies based on the task distribution. A key method in this area is model-agnostic meta-learning (MAML), which optimizes a model's initial parameters so it can adapt quickly to new tasks using only a small amount of training data [22]. Building on this, Lee et al. [23] proposed MetaOptNet, a method that uses linear classifiers as base learners. By applying implicit differentiation and dual optimization techniques, MetaOptNet effectively bal-

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ances high-dimensional feature representation with classification performance, while maintaining low computational cost.

2.1.3 | Data Augmentation-Based Methods

Data augmentation techniques expand the training datasets to overcome data scarcity [24]. For instance, Zhang et al. [25] employed generative adversarial networks (GANs) to generate artificial images for few-shot classes, while Li et al. [8] proposed adversarial feature hallucination to directly generate diverse feature vectors. Recent work by Bär et al. [7] proposed frozen feature augmentation (FroFA), a method that adapts image-style augmentations to frozen vision transformer features by scaling and transforming them in a stylistically consistent manner.

2.1.4 | Multi-Modal-Based Methods

Multi-modal methods leverage intermodal complementarity to enhance feature representations and classification performance through joint embedding optimization. Radford et al. [12] established foundational work with CLIP, employing contrastive learning to align visual and textual embeddings through large-scale image-text pretraining, enabling effective zero-shot transfer via semantic consistency. The field evolved with unified foundation models like Florence [26], which introduced dynamic fusion mechanisms for integrating visual, linguistic, and spatial features through task-agnostic representation learning. Recent architectural innovations exemplified by ImageBind [27] extended multi-modal alignment to diverse sensory inputs through shared semantic spaces, enabling unsupervised learning across six modalities. Concurrently, parameter-efficient adaptation methods have emerged as critical solutions for cross-modal transfer, utilizing lightweight modules to maintain performance while substantially reducing computational overhead. More recent studies have extended medical few-shot learning toward domain-aware vision–language adaptation. PM^2 introduced a prompting multi-modal paradigm for few-shot medical image classification, where text prompts are used as supplementary semantic inputs to enhance image representations [13]. In domain-specific medical settings, FLAIR incorporated expert textual supervision for retinal image understanding and showed strong transferability and strong zero-shot generalization in retinal tasks [28], while TITAN demonstrated that large-scale multimodal pretraining on pathology reports and synthetic captions can improve pathology foundation modeling across settings including few-shot and zero-shot classification [29]. For pathological image analysis, PATE further enhanced CLIP adaptation by introducing prompt-based text-image embedding adaptation for few-shot pathological image classification [30]. These methodological advances systematically address fundamental challenges in modality heterogeneity while establishing theoretical foundations for joint embedding space optimization.

2.2 | Prompt Learning

Prompt learning is a powerful methodology for transferring knowledge from large pre-trained models to small datasets [31]. By designing prompts tailored to downstream tasks, prompt learning enables efficient adaptation of pre-trained models while improving generalization to new tasks or classes. Recent surveys have further shown that prompt engineering has become a central paradigm in vision–language model adaptation, especially in low-resource and parameter-efficient settings [32]. Schick and Schütze pioneered pattern-exploiting training (PET), which combines predefined discrete templates with knowledge distillation [33]. Later, prefix-tuning introduced continuous prompts and enabled more flexible and parameter-efficient optimization [34]. In the medical domain, recent surveys have also highlighted the growing importance of language-guided adaptation for visual recognition and reasoning tasks [35]. Recent research has further advanced prompt learning across multimodal domains. For example, multi-modal prompt learning (MaPLe) leverages visual-to-language couplings in CLIP to improve the alignment between the two modalities and achieves 3.45% higher accuracy than previous techniques on unseen classes [36]. More recently, PM^2 introduced a prompting-based multimodal framework for few-shot medical image classification by incorporating supplementary text prompts as semantic guidance [13], while PATE enhanced CLIP adaptation for few-shot pathological image classification through deep learnable prompts in both vision and language encoders [30].

In addition, graph structure prompt learning (GPL) has introduced task-agnostic structural losses into graph neural networks (GNNs) in order to improve node and graph representation performance, achieving up to 24.15% improvement over baseline GNN performance when predicting edges [37].

Prompt learning can be used for a wide range of applications that have successfully addressed challenges in few-shot and zero-shot learning, domain adaptation, and computational efficiency. For example, the zero-shot sample for GPT-3 is generated using the in-context learning framework, which translates tasks into natural language, allowing the model to be given an entire task description when generating samples [38]. In instances where it is necessary to avoid overfitting to the task in low-resource environments, methods such as RETROPROMPT develop a means to dynamically extract the relevant context from training data by separating memorisation of training data from adapting to the task [39]. There are still many important issues that need to be resolved, and many of the most common prompt-sensitivity issues have become more technical, as small variations in the template cause model performance to become unstable. Continuous prompts may be efficient in terms of parameter count, but they have the disadvantage of reducing interpretability [34]; additionally, many prompts can be computationally intensive, necessitating methods to compress resources, such as knowledge distillation. Future work will become increasingly focused on automating the generation of prompts through the use of meta-learning, increasing the use of multimodal integration, and implementing ethical safeguards to limit bias in prompt outputs; in doing so, the overall goal will be to improve prompt learning systems' performance.

3 | Method

For clarity, we use bold lower-case letters for vectors, bold upper-case letters for matrices, superscript i for sample indices, and subscript c for class indices throughout the paper.

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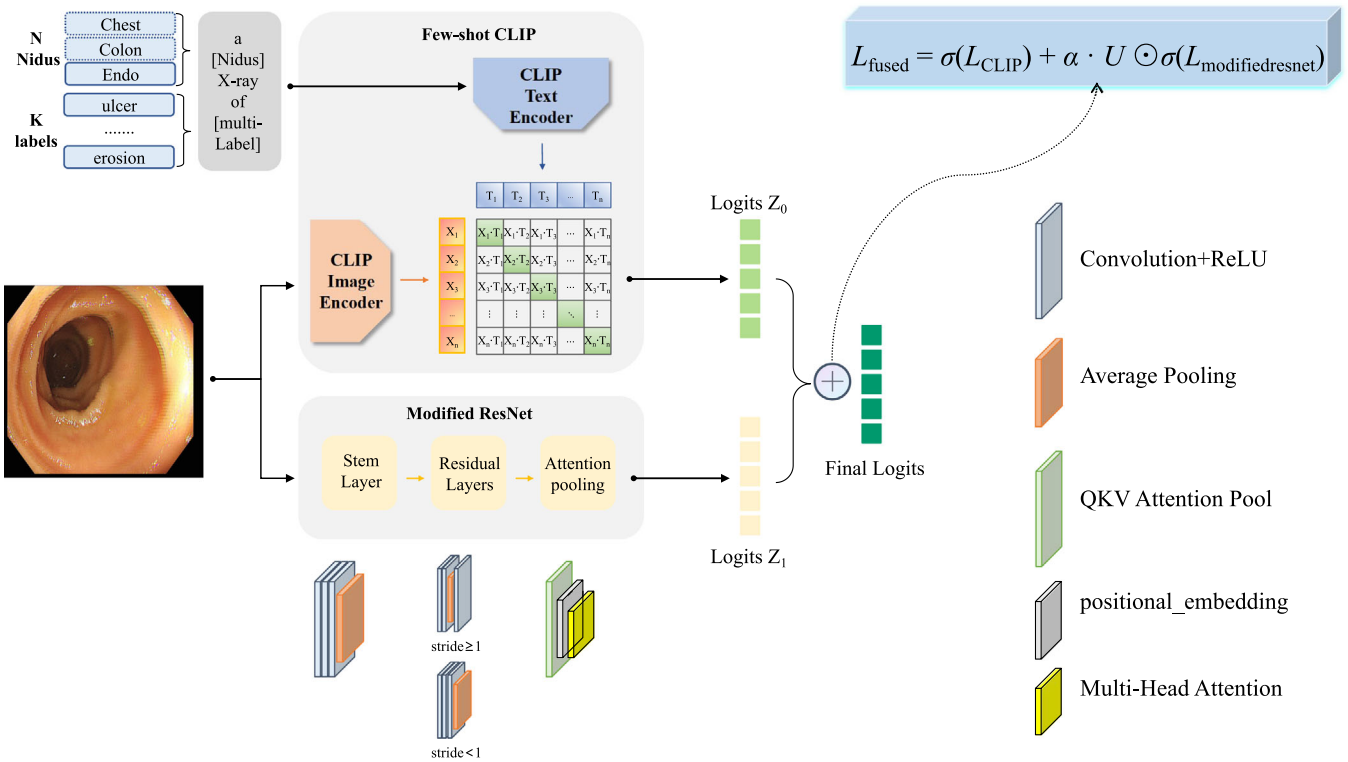


FIGURE 2 | The overall framework of the proposed PKDNet. The framework consists of a semantic alignment branch based on CLIP and a structural perception branch based on an enhanced ResNet. An uncertainty-aware fusion mechanism is introduced to integrate complementary semantic and structural information for robust medical image classification.

3.1 | Problem Formulation

Let $\mathcal{D} = \{(\mathbf{I}^i, \mathbf{y}^i)\}_{i=1}^N$ denote a medical image dataset, where $\mathbf{I}^i \in \mathbb{R}^{3\times H\times W}$ is the i-th medical image and $\mathbf{y}^i \in \{0,1\}^C$ is the corresponding label vector over C disease categories. Here, N denotes the number of samples and C denotes the number of classes. The goal is to learn a discriminative mapping

$$\mathcal{F} : \mathbb{R}^{3\times H\times W} \to \mathbb{R}^C, \tag{1}$$

which predicts category scores for each input image.

In few-shot medical image classification, disease semantics are often expressed at the lesion or organ level, whereas discriminative visual evidence usually appears as localized structural variations, such as subtle texture irregularities, blurry boundaries, and morphological deformations. This motivates a dual-branch formulation that explicitly models semantic–structural complementarity. As illustrated in Figure 2, the proposed PKDNet consists of two parallel branches: (i) a CLIP-based semantic alignment branch and (ii) an enhanced ResNet-based structural perception branch. The outputs of the two branches are further integrated by an uncertainty-aware fusion mechanism.

Unless otherwise specified, bold lower-case symbols denote vectors, bold upper-case symbols denote matrices, superscript i indexes samples, superscript (l) indexes network stages, and subscript c denotes the c-th category.

3.2 | Branch-Specific Preprocessing

Given an input image $\mathbf{I}^i \in \mathbb{R}^{3\times 224\times 224}$, branch-specific normalization is applied before feeding it into the two branches:

$$\mathbf{I}^i_{\text{clip}} = \frac{\mathbf{I}^i - \boldsymbol{\mu}_{\text{clip}}}{\boldsymbol{\sigma}_{\text{clip}}}, \qquad \mathbf{I}^i_{\text{ert}} = \frac{\mathbf{I}^i - \boldsymbol{\mu}_{\text{ert}}}{\boldsymbol{\sigma}_{\text{ert}}}, \tag{2}$$

where $(\boldsymbol{\mu}_{\text{clip}}, \boldsymbol{\sigma}_{\text{clip}})$ are the normalization statistics used in CLIP pretraining, and $(\boldsymbol{\mu}_{\text{ert}}, \boldsymbol{\sigma}_{\text{ert}})$ correspond to the ImageNet normalization used by the enhanced ResNet branch.

This decoupled preprocessing preserves the feature distributions expected by the two pretrained encoders and ensures stable optimization.

3.3 | Semantic Feature Extraction via CLIP

To provide lesion-level semantic guidance, PKDNet leverages the vision–language alignment capability of CLIP.

For each class $c \in \{1, \ldots, C\}$, a learnable prompt is constructed as

$$t_c = [\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_m, e_c], \tag{3}$$

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where $\mathbf{T}_j \in \mathbb{R}^{d_t}$ denotes the j-th learnable context token, m is the number of context tokens, and e_c is the embedding of the class name of category c.

The text encoder $g_\phi(\cdot)$ maps each prompt into the shared embedding space:

$$\mathbf{w}_c = \frac{g_\phi(t_c)}{\|g_\phi(t_c)\|_2} \in \mathbb{R}^d. \tag{4}$$

Stacking all class text embeddings yields the semantic prototype matrix

$$\mathbf{W}_{\text{CLIP}} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_C] \in \mathbb{R}^{d\times C}. \tag{5}$$

For the i-th image, the CLIP image encoder $f_\theta(\cdot)$ produces a normalized semantic feature

$$\mathbf{f}^i_{\text{CLIP}} = \frac{f_\theta(\mathbf{I}^i_{\text{CLIP}})}{\|f_\theta(\mathbf{I}^i_{\text{CLIP}})\|_2} \in \mathbb{R}^d. \tag{6}$$

The similarity between image i and class c is computed as

$$s_{ic} = \frac{(\mathbf{f}^i_{\text{CLIP}})^\top \mathbf{w}_c}{\tau}, \tag{7}$$

where τ is a temperature parameter.

For a mini-batch of size B, the semantic feature matrix is

$$\mathbf{F}_{\text{CLIP}} = [\mathbf{f}^1_{\text{CLIP}}, \mathbf{f}^2_{\text{CLIP}}, \ldots, \mathbf{f}^B_{\text{CLIP}}]^\top \in \mathbb{R}^{B\times d}, \tag{8}$$

and the corresponding similarity matrix is $\mathbf{S} \in \mathbb{R}^{B\times C}$.

To optimize semantic alignment, a symmetric contrastive objective is employed:

$$\mathcal{L}_{\text{image}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(s_{i,y_i})}{\sum_{c=1}^{C}\exp(s_{ic})}, \tag{9}$$

$$\mathcal{L}_{\text{text}} = -\frac{1}{C}\sum_{c=1}^{C} \log \frac{\exp\left(\frac{1}{N_c}\sum_{i\in S_c} s_{ic}\right)}{\sum_{i=1}^{B}\exp(s_{ic})}, \tag{10}$$

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2}\left(\mathcal{L}_{\text{image}} + \mathcal{L}_{\text{text}}\right), \tag{11}$$

where S_c denotes the set of training samples belonging to class c, and $N_c = |S_c|$ is the number of such samples.

This formulation enables CLIP to capture high-level disease semantics, while its limited sensitivity to fine-grained structure motivates complementary structural modeling.

Specifically, we set the context length to $m = 16$, following standard configurations in continuous prompt learning to maintain parameter efficiency. Furthermore, randomly initializing prompt tokens may lead to less stable optimization in low-data regimes. To provide a stable starting point, we adopt a hybrid initialization strategy. Instead of a purely random initialization, the initial tokens are embedded using dataset-specific base templates (e.g., "a chest X-ray showing signs of") to provide domain-relevant semantic cues. The remaining tokens are initialized from a zero-mean Gaussian distribution $\mathcal{N}(0, 0.02^2)$. This structured initialization provides a stable starting point for learning under limited supervision. During training, the CLIP image encoder and text encoder are kept frozen, and only the learnable context vectors and task-specific components are optimized.

3.4 | Structural Feature Extraction via Enhanced ResNet

To complement semantic abstraction, PKDNet employs an enhanced ResNet branch (illustrated in Figure 3) to capture fine-grained structural information.

Given $\mathbf{I}^i_{\text{ert}}$, the feature extraction pipeline is defined as

$$\mathbf{F}^{(0),i}_{\text{ert}} = \text{Stem}(\mathbf{I}^i_{\text{ert}}), \tag{12}$$

$$\mathbf{F}^{(l),i}_{\text{ert}} = \mathcal{B}_l\left(\mathbf{F}^{(l-1),i}_{\text{ert}}\right), \quad l \in \{1, \ldots, 4\}, \tag{13}$$

where $\text{Stem}(\cdot)$ denotes the modified convolutional stem and $\mathcal{B}_l(\cdot)$ denotes the l-th residual stage.

Let $\mathbf{X}^i \in \mathbb{R}^{M\times d}$ denote the flattened spatial feature map from the last stage, where M is the number of spatial tokens and d is the channel dimension. To preserve spatial sensitivity, global average pooling is replaced with an attention-based pooling mechanism:

$$\text{AttnPool}(\mathbf{X}^i) = \text{softmax}\left(\frac{\mathbf{Q}^i(\mathbf{K}^i)^\top}{\sqrt{d}}\right)\mathbf{V}^i, \tag{14}$$

where

$$\mathbf{Q}^i = \mathbf{X}^i\mathbf{W}_Q, \qquad \mathbf{K}^i = \mathbf{X}^i\mathbf{W}_K, \qquad \mathbf{V}^i = \mathbf{X}^i\mathbf{W}_V. \tag{15}$$

The resulting structural representation is

$$\mathbf{f}^i_{\text{ert}} = \text{AttnPool}\left(\mathbf{F}^{(4),i}_{\text{ert}}\right) \in \mathbb{R}^d. \tag{16}$$

For a mini-batch of size B, the structural feature matrix is

$$\mathbf{F}_{\text{ert}} = [\mathbf{f}^1_{\text{ert}}, \mathbf{f}^2_{\text{ert}}, \ldots, \mathbf{f}^B_{\text{ert}}]^\top \in \mathbb{R}^{B\times d}. \tag{17}$$

This branch emphasizes local texture and boundary cues, complementing the lesion-level semantic focus of CLIP.

3.5 | Logits Generation

Due to the heterogeneous nature of semantic and structural representations, PKDNet adopts branch-specific logit formulations.

The CLIP-based logits are computed as

$$\mathbf{Z}_{\text{CLIP}} = \gamma\,\mathbf{F}_{\text{CLIP}}\mathbf{W}_{\text{CLIP}} \in \mathbb{R}^{B\times C}, \tag{18}$$

where γ is the logit scaling factor. In our implementation, $\gamma = 100$.

For the enhanced ResNet branch, a prototype-based classifier is employed. Specifically, the class prototype of category c is

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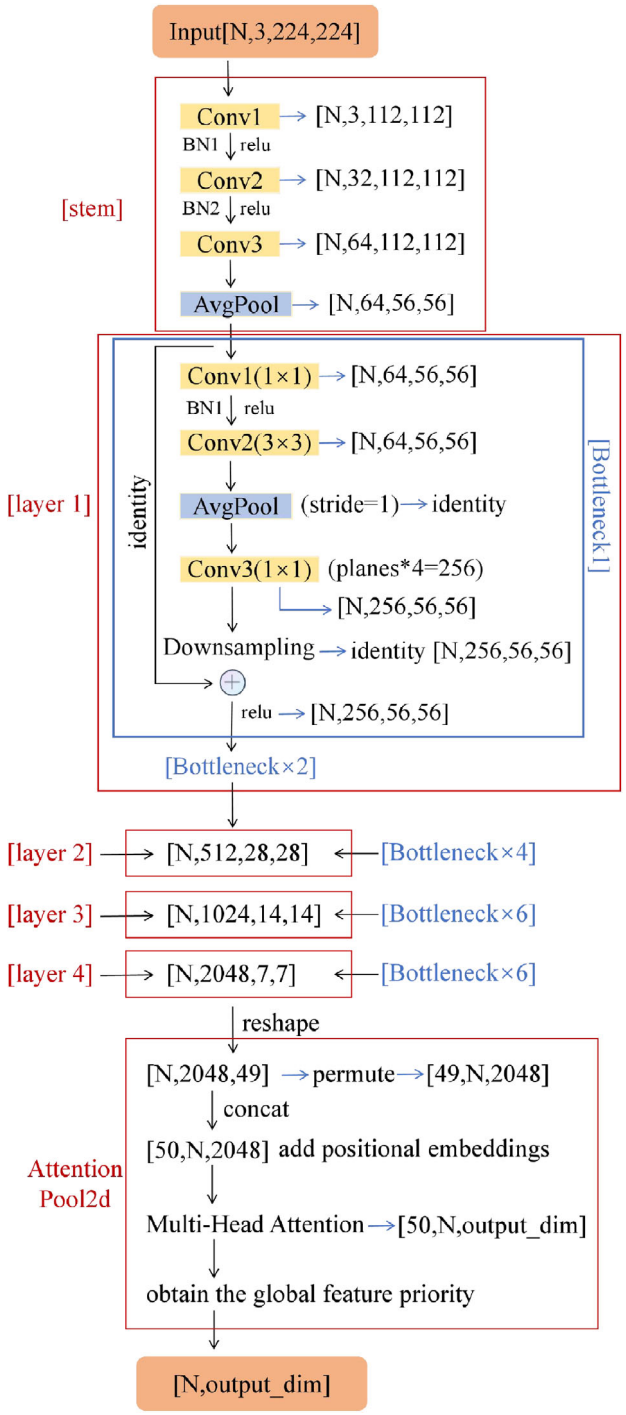


FIGURE 3 | Architecture of the enhanced ResNet branch, illustrating the modified convolutional stem, residual bottleneck blocks, and attention-based pooling mechanism for preserving fine-grained structural information.

initialized as

$$\mathbf{p}_c = \frac{1}{N_c} \sum_{i \in S_c} \mathbf{f}_{\text{ert}}^i \in \mathbb{R}^d. \tag{19}$$

Stacking all structural prototypes yields

$$\mathbf{W}_{\text{ert}} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_C] \in \mathbb{R}^{d \times C}. \tag{20}$$

The corresponding structural logits are then computed as

$$\mathbf{Z}_{\text{ert}} = \mathbf{F}_{\text{ert}} \mathbf{W}_{\text{ert}} \in \mathbb{R}^{B \times C}. \tag{21}$$

This heterogeneous modeling preserves semantic similarity in the CLIP branch while maintaining discriminative structural boundaries in the enhanced ResNet branch.

It should be emphasized that PKDNet does not directly align or add the raw feature embeddings produced by the two branches. Instead, each branch is first mapped into the shared class-wise prediction space $\mathbb{R}^C$. Both $\mathbf{Z}_{\text{CLIP}}$ and $\mathbf{Z}_{\text{ert}}$ have the same category dimension, and the c-th output dimension corresponds to the same disease category c. Therefore, the heterogeneous semantic and structural representations are aligned at the prediction level before fusion.

3.6 | Uncertainty-Aware Fusion

To dynamically balance semantic and structural predictions, PKDNet introduces an uncertainty-aware fusion mechanism. Based on the prediction-level alignment described above, the fusion is performed in the shared class-wise prediction space rather than in the raw heterogeneous feature spaces.

For the i-th sample, the CLIP class-wise confidence score is first computed as

$$p_{\text{CLIP}}^i = \sigma\left(z_{\text{CLIP}}^i\right) \in [0,1]^C, \tag{22}$$

where $z_{\text{CLIP}}^i \in \mathbb{R}^C$ denotes the i-th row of Z_{CLIP}, and $\sigma(\cdot)$ denotes the element-wise sigmoid function.

Similarly, the structural confidence score of the enhanced ResNet branch is computed as

$$p_{\text{ert}}^i = \sigma\left(z_{\text{ert}}^i\right) \in [0,1]^C, \tag{23}$$

where $z_{\text{ert}}^i \in \mathbb{R}^C$ denotes the i-th row of Z_{ert}.

Semantic uncertainty is quantified using an entropy-based heuristic:

$$h_i = -\sum_{c=1}^{C} p_{\text{CLIP},c}^i \log p_{\text{CLIP},c}^i. \tag{24}$$

An uncertainty weight is defined as

$$u_i = \exp\left(\beta \frac{h_i}{\log C}\right), \tag{25}$$

where $\beta > 0$ controls uncertainty sensitivity. The exponential mapping provides a smooth and monotonic amplification of uncertainty, such that structurally informative cues are increasingly emphasized when the semantic branch becomes less confident.

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The fused prediction for sample i is computed using uncertainty-aware normalized additive fusion:

$$p^i_{\text{fused}} = \frac{p^i_{\text{CLIP}} + \alpha u_i p^i_{\text{ert}}}{1 + \alpha u_i} \tag{26}$$

where $\alpha > 0$ controls the overall contribution of the structural branch, and the scalar u_i is broadcast to all categories of sample i.

The mathematical rationale for this formulation rests on scale calibration and probabilistic ensembling. The two branches generate logits through different mechanisms: the CLIP logits are obtained from scaled image-text similarities, whereas the enhanced ResNet logits are computed from prototype similarities. Therefore, their raw logit distributions may have different numerical scales. The sigmoid function maps these logits into bounded class-wise confidence scores, making them comparable in the shared label space. The normalized additive fusion can be interpreted as an uncertainty-weighted linear opinion pool, with effective weights $1/(1 + \alpha u_i)$ and $\alpha u_i/(1 + \alpha u_i)$ for the semantic and structural branches, respectively. Since these weights are non-negative and sum to one, p^i_{fused} remains in $[0, 1]^C$ and is compatible with BCE supervision. This parameter-free fusion also avoids introducing additional trainable fusion modules, which helps reduce overfitting risk under few-shot supervision.

For a mini-batch, the fused predictions form $P_{\text{fused}} \in \mathbb{R}^{B\times C}$.

This formulation increases reliance on structural cues when semantic predictions are uncertain, mitigating semantic overconfidence.

3.7 | Loss Function

To jointly optimize semantic alignment and structural discrimination while preventing fusion collapse, a dual-supervision strategy is adopted.

The fusion loss is defined as

$$\mathcal{L}_{\text{fuse}} = \frac{1}{B \cdot C} \sum_{i=1}^{B} \sum_{c=1}^{C} \text{BCE}\left(p^i_{\text{fused},c}, y^i_c\right), \tag{27}$$

where $p^i_{\text{fused},c}$ denotes the fused prediction probability of sample i for category c, and y^i_c is the corresponding binary target.

An auxiliary structural loss is applied as

$$\mathcal{L}_{\text{ert}} = \frac{1}{B \cdot C} \sum_{i=1}^{B} \sum_{c=1}^{C} \text{BCE}\left(p^i_{\text{ert},c}, y^i_c\right). \tag{28}$$

The overall optimization objective is

$$\mathcal{L}_{\text{total}} = \lambda \mathcal{L}_{\text{fuse}} + (1 - \lambda)\mathcal{L}_{\text{ert}}, \tag{29}$$

where $\lambda \in [0, 1]$ balances the fused supervision and the auxiliary structural supervision.

The auxiliary structural supervision preserves the discriminative capability of the enhanced ResNet branch, while the fused supervision encourages effective semantic–structural integration. This dual supervision prevents degenerate solutions in which one branch dominates the fusion process.

TABLE 1 | Three medical datasets (ChestDR, Colon and Endo).

Dataset	Training	Validation	Testing
ChestDR	2140	2708	3869
Colon	5654	4355	7651
Endo	1810	2055	2936

4 | Experiments

4.1 | Datasets

As shown in Table 1, to evaluate the performance of the proposed method, the MedFMC2023 grand challenge public classification dataset [40] was used. It covers three different diseases:

Thoracic Disease Screening (ChestDR) [40]. ChestDR is a thoracic disease screening dataset composed of X-ray images collected from regional hospitals in Hubei and Jiangxi Provinces, China. A total of 4848 frontal radiography images (one per patient) are included, covering 19 common thoracic abnormalities: pleural_effusion, nodule, pneumonia, cardiomegaly, hilar_enlargement, fracture_old, fibrosis, aortic_calcification, tortuous_aorta, thickened_pleura, TB, pneumothorax, emphysema, atelectasis, calcification, pulmonary_edema, increased_lung_markings, elevated_diaphragm, and consolidation.

Pathological Tumor Tissue Classification (Colon) [41]. Colon originates from the DigestPath 2019 Challenge, curated by the Histo Pathology Diagnostic Center. It contains tissue patches extracted from whole-slide images of 396 patients undergoing colonoscopy examinations (2017–2019). Patches are categorized into binary classes: Normal tissue and lesion tissue. Initial labels are generated algorithmically based on segmentation masks, then validated by senior pathologists.

Endoscopy Images for Lesion Classification (Endo) [40]. The Endo dataset recorded during colonoscopy examinations on workstations at Renji Hospital, School of Medicine, Shanghai Jiao Tong University. These images were collected from 80 patients and cover four types of lesions: ulcer, erosion, polyp, and tumor. Labels were initially annotated by junior gastroenterologist using health records and reports, then verified by senior gastroenterologist with over 10 years of experience.

4.2 | Comparative Methods

To demonstrate the effectiveness of our proposed method, this paper utilizes the publicly methods on three datasets mentioned above for comparative experiments.

The details of the comparison method are as follows.

- Full [19]: Unfreezes the entire backbone for end-to-end parameter tuning.

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- Linear [19]: Freezes the backbone and trains only the top classification head, serving as the parameter-efficient baseline.
- MLP-3 [42]: A method that inserts a three-layer lightweight MLP module before the classification head, tuning exclusively this component.
- VP [43]: A method that adds learnable prompt vectors directly to input image channels, creating adaptive input perturbations that shift pixel-level distributions.
- VPT [44]: The conventional prompt tuning approach that prepends learnable prompts to input token sequences, freezing the backbone while tuning only prompt parameters.
- EPT [42]: A novel prompt tuning paradigm that embeds learnable prompts within channel-expanded dimensions of the attention mechanism. It optimizes token representations through patch-wise scaling operations applied to the attention logits (pre-softmax scores), preserving original information while dynamically calibrating feature distributions.

4.3 | Experimental Setting

The research is conducted using the standard *N-way K-shot* FSL paradigm, where each task consists of N (number of categories) diseases and K (number of support samples) labeled support samples. In this research, we focus on the difficult cross-domain FSL scenarios $K = 1$, $K = 5$, and $K = 10$, which correspond to very little, moderate, and relatively high supervision levels, respectively. An important distinction is that $K = 1$ refers to a single support sample per class in the N-way task, not a single aggregated image.

The backbone of the model is initialized with ImageNet-21k pretraining weights. This creates a very large domain gap between the source domain (natural images) and the target domain (medical images). Therefore, the setup accurately assesses a model's ability to generalize across domains in few-shot settings, an especially relevant scenario in medical imaging given the limited amount of annotated data available. The experiments will include three medical imaging datasets, the Chest, Colon, and Endo databases, and these datasets consist of both multiclass and multilabel classification tasks.

The chosen evaluation metrics are tailored to the characteristics of each dataset. For the Colon dataset, which will have multiclass classification, the evaluation metric will be the total classification accuracy (Acc). mAP has been selected as the metric to evaluate the performance of both the Chest and Endo datasets because it provides a better understanding of class-specific performance when class frequencies are imbalanced. All datasets will be reported in the 1-shot, 5-shot, and 10-shot configurations to evaluate the models' robustness across levels of supervision.

Moreover, to provide a global view of the models' performance across datasets and shot settings (1-shot, 5-shot, and 10-shot), we will also report how much better fine-tuning-based approaches perform compared to linear baselines. These performance improvements will reflect the success of adaptive representation learning under limited supervision and will be achieved without relying on fully supervised methods for model training.

4.4 | Training Dynamics and Optimization Analysis

To analyze the optimization behavior and training stability of the proposed model, Figure 4 illustrates the evolution of the training loss together with the corresponding learning rate schedule across epochs. The monotonic decrease in training loss during the early training stage indicates effective gradient-based optimization under the adopted learning rate warmup and cosine decay strategy.

Specifically, the training process has three distinct phases. Between epochs 1 and 15, the loss shows a sharp decline, suggesting rapid adaptation of model parameters and efficient learning of dominant low-frequency patterns in the feature space. This behaviour is consistent with the use of a relatively high learning rate during early optimisation. In the middle phase (epochs 15–30), the loss curve enters a transitional plateau, reflecting a gradual refinement of representations as the optimiser explores flatter regions of the parameter landscape. Finally, in the last stage, the loss value converges steadily, fluctuating between 0.755 and 0.756. This indicates that the convergence process is stable when the learning rate is reduced.

This progression shows that the effective convergence of an optimization approach that provides effective convergence without oscillation or divergence is a key element to a model's quantitative performance.

4.5 | Ablation Study

To assess the contribution of CLIP to the proposed framework, we conduct an ablation study by removing the CLIP module while keeping all other components unchanged. This variant is denoted as *without_CLIP*.

As shown in Figure 5, removing CLIP results in consistent and significant reductions in performance across all evaluation metrics and datasets. This drop in performance shows that when we remove CLIP from the model, it has a very weak ability to produce strong semantic representations, ultimately resulting in reduced discriminative ability in a cross-domain few-shot setting. In the 5-shot setting, the proposed model achieves 77.8% improvement over the *without_CLIP* model on the chest dataset, 91.3% on the endoscopy dataset, and 34.4% on the colon dataset. These large performance differences indicate that CLIP is highly important because it enables the model to establish higher-level semantic alignment with the structural features extracted from the auxiliary backbones. CLIP's semantic guidance is needed for robust generalizable performance in the proposed fusion framework.

4.5.1 | Effect of Fusion Strategy

To further validate the effectiveness of the proposed uncertainty-aware fusion mechanism, we compare it with several alternative fusion strategies under the 5-shot setting, including modified ResNet-only, CLIP-only, and simple linear fusion. As shown in Table 2, the proposed method consistently achieves the best performance on all three datasets.

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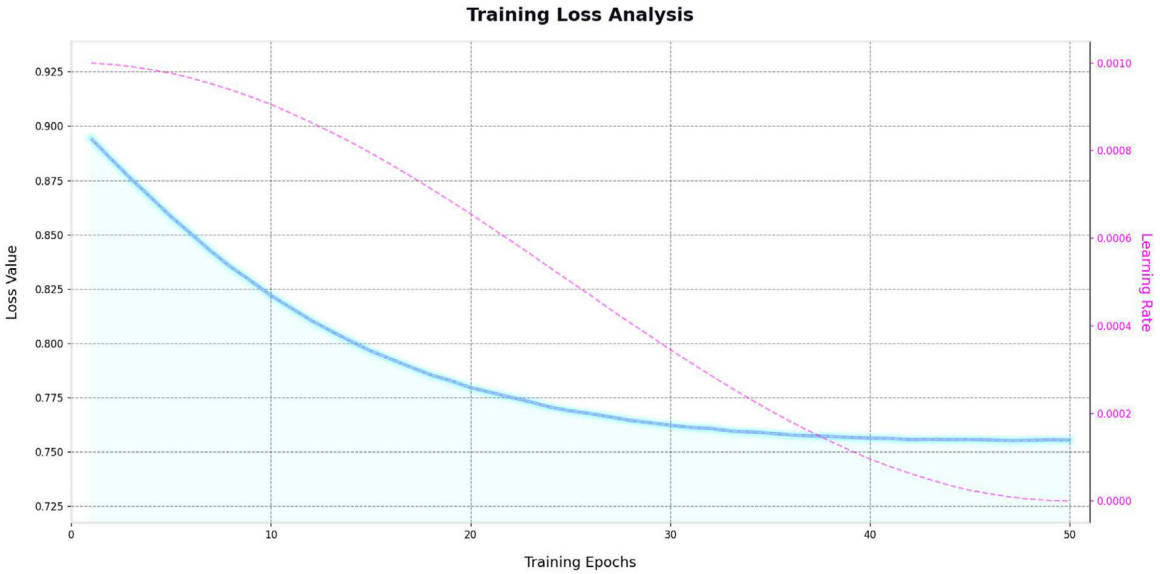


FIGURE 4 | Training loss curve and learning rate schedule over epochs.

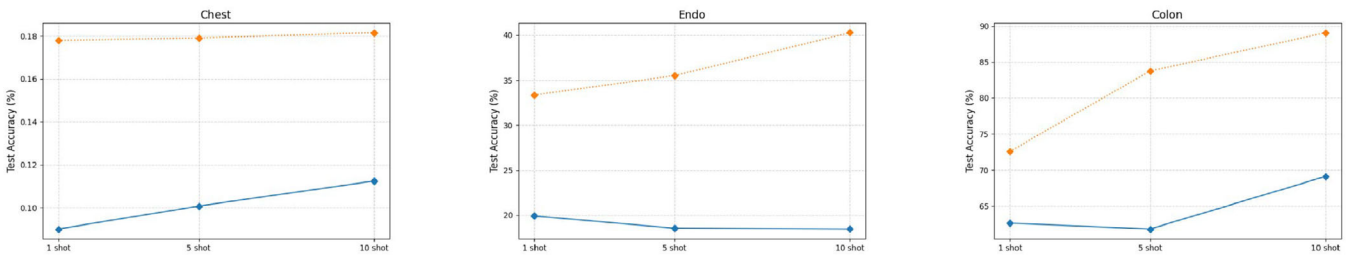


FIGURE 5 | Ablation study on the effect of CLIP integration. Performance comparison between PKDNet and its variant without CLIP (*without_CLIP*) on the Chest, Endo, and Colon datasets under few-shot learning settings, demonstrating the contribution of CLIP to semantic representation and overall performance.

TABLE 2 | Ablation study of different fusion strategies under the 5-shot setting. "Modified ResNet-only" denotes the structural branch without CLIP guidance. "Ours" refers to the proposed uncertainty-aware fusion mechanism. The last column reports the average performance across the three datasets.

Method	ChestDR (mAP)	Endo (mAP)	Colon (Acc)	Avg.
Modified ResNet-only (without CLIP)	10.07	18.56	62.33	41.78
CLIP-only	16.84	32.41	81.96	43.74
CLIP + modified ResNet (linear)	17.42	34.63	83.18	45.08
CLIP + modified ResNet (Ours)	**17.91**	**35.51**	**83.77**	**45.73**

Compared with CLIP-only, the proposed dual-stream fusion improves the average performance from 43.74 to 45.73, indicating that structural cues provide complementary fine-grained information beyond semantic alignment. Moreover, our uncertainty-aware fusion also outperforms simple linear fusion, demonstrating that adaptive sample-wise reweighting is more effective than static fusion. These results validate that the performance gain of PKDNet is not merely due to increased model capacity, but arises from a principled fusion strategy that effectively integrates semantic priors with structural cues.

4.5.2 | Effect of Pooling Strategy

To quantify the contribution of the proposed attention-based pooling module, we compare it with two standard alternatives, namely global max pooling (GMP) and global average pooling (GAP). As shown in Table 3, attention pooling consistently out-

TABLE 3 | Ablation study of different pooling strategies in the structural branch under the 5-shot setting. Attention pooling consistently achieves the best performance across all datasets.

Pooling type	ChestDR (mAP)	Endo (mAP)	Colon (Acc)
Global max pooling (GMP)	16.72	33.08	82.61
Global average pooling (GAP)	17.03	33.84	83.02
Attention pooling	**17.54**	**34.78**	**83.53**

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TABLE 4 | Stepwise ablation of the enhanced ResNet branch under the 5-shot setting.

Variant	Deep stem	Anti-aliasing	Attention pooling	ChestDR (mAP)	Endo (mAP)	Colon (Acc)
Baseline ResNet-style	✗	✗	✗	15.81	32.22	82.47
+ Deep stem	✓	✗	✗	16.72	33.63	82.95
+ Deep stem + anti-aliasing	✓	✓	✗	17.30	34.94	83.28
Full modified ResNet (ours)	✓	✓	✓	**18.11**	**35.75**	**83.66**

performs both GMP and GAP on all three datasets. Specifically, it improves performance from 17.03 to 17.54 mAP on ChestDR and from 33.84 to 34.78 mAP on Endo.

This improvement can be attributed to the ability of attention pooling to selectively focus on diagnostically relevant regions, rather than aggregating features uniformly or relying on extreme activations. This is particularly important in medical imaging, where subtle local patterns are critical for classification.

4.5.3 | Component-Wise Ablation of the Enhanced ResNet Branch

As shown in Table 4, each architectural modification contributes positively and cumulatively improves performance. The deep stem improves early feature extraction by replacing the large-kernel convolution with a sequence of smaller convolutions, which is particularly beneficial for capturing fine-grained structures in medical images.

Incorporating anti-aliasing further enhances performance by alleviating the aliasing effect introduced by strided convolutions, leading to more stable feature representations. Notably, attention pooling brings the largest performance gain, suggesting that adaptively focusing on discriminative regions is crucial in few-shot scenarios where global pooling may dilute critical local patterns.

4.6 | Quantitative Analysis

Table 5 reports the quantitative performance of different methods under 1-shot, 5-shot, and 10-shot settings on the ChestDR, Endo, and Colon datasets. The proposed PKDNet achieves the highest overall average accuracy of 45.16%, outperforming the Linear baseline by 20.49% and the strongest competing method EPT by 13.41%, demonstrating a clear advantage under cross-domain few-shot conditions. On the ChestDR multilabel dataset, regardless of the shot configuration, PKDNet outperforms other methods across all configurations. PKDNet achieves 17.80% mAP in the extreme low-shot condition, which is significantly higher than EPT's 12.73% and the linear baseline's 11.93%. In this configuration, PKDNet achieves a large absolute gain (5.07%) over EPT and (5.87%) over the linear baseline. This performance gap also indicates PKDNet's stability, as evidenced by its 18.15% mAP under the 10-shot configuration. Therefore, it is evident that PKDNet is capable of stable semantic-structural generalization, not just performance improvement, which is a function of increased supervision. The Endo dataset exhibits the most pronounced performance gains. Under the settings of 1-shot, 5-shot and 10-shot, PKDNet achieved mAP values of 33.36%, 35.51% and 40.28%, respectively. Compared with the strongest baseline method EPT, PKDNet demonstrated a performance improvement of 15.70% (33.36% vs. 17.66%) in the 1-shot setting and 15.21% (40.28% vs. 25.07%) in the 10-shot setting. Such a significant performance advantage indicates that explicitly integrating lesion-level semantic guidance with fine-grained structural cues is particularly effective in endoscopy image analysis tasks, as it is difficult to distinguish visually similar lesion categories under limited supervision conditions in such tasks. On the Colon dataset, which involves multiclass classification, PKDNet achieves competitive performance against the best-performing baselines. While EPT attains the highest accuracy in the 1-shot setting (75.31%), PKDNet surpasses all methods in the 5-shot and 10-shot settings, achieving 83.77% and 89.09% accuracy, respectively. The relative improvement achieved under a higher sample size indicates that PKDNet can more effectively utilize the additional supervision information by jointly optimizing semantic alignment and structural discrimination capabilities, thereby enhancing the class separability. In all datasets, the methods based on prompts and adapters (VPT and EPT) consistently outperformed the linear baseline, which confirmed the effectiveness of the parameter-efficient adaptation strategy in cross-domain small-sample learning. However, the average performance of the full fine-tuning strategy was 9.34% lower than the linear baseline, highlighting the risk of overfitting when updating all parameters under extremely limited supervision conditions. In contrast, PKDNet maintained stable performance across different datasets and sample sizes, demonstrating a good balance between adaptation flexibility and generalization stability.

4.7 | Complexity and Efficiency Analysis

We further analyze the computational complexity and efficiency of PKDNet, as summarized in Table 6. Compared with the parameter-efficient baseline EPT, PKDNet introduces a dual-stream architecture, resulting in an increased total parameter count (173.16M vs. 85.81M). However, it is important to note that most of these parameters are inherited from pretrained models and remain frozen during training. Consequently, PKDNet requires fewer trainable parameters than EPT (1.45M vs. 1.94M), highlighting its efficiency in adapting large-scale pretrained knowledge to few-shot medical tasks. In terms of computational cost, PKDNet incurs only a moderate increase in FLOPs (18.76G vs. 15.40G) and training time (62.37s vs. 56.18s), which is acceptable given the improved representational capacity brought by the dual-stream design. Considering the substantial performance gains reported in Table 6, the additional computational overhead

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TABLE 5 | Quantitative comparison under cross-domain few-shot settings. Performance comparison of different methods on the ChestDR, Endo, and Colon datasets under 1-shot, 5-shot, and 10-shot learning configurations. Accuracy is reported for the Colon dataset, while mean average precision (mAP) is reported for ChestDR and Endo. The last column shows the average performance across all datasets and shot settings, with relative improvements over the Linear baseline shown in parentheses.

	ChestDR			Endo			Colon			
	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	1-shot	5-shot	10-shot	average
Full [19]	10.61	11.16	11.99	14.05	15.66	17.57	63.90	77.02	83.85	33.98(-9.34%)
Linear [19]	11.93	14.88	16.75	15.60	20.18	21.30	70.24	81.22	85.23	37.48(0.00%)
MLP-3 [42]	12.54	15.85	17.78	16.34	20.29	21.67	70.56	81.63	86.13	38.09(+1.63%)
VP [43]	12.22	15.65	17.33	15.67	19.46	20.79	73.70	81.26	85.11	37.91(+1.15%)
VPT [44]	12.15	14.88	17.54	17.60	21.27	22.89	74.44	82.15	86.25	38.79(+3.50%)
EPT [42]	12.73	16.16	18.09	17.66	22.8	25.07	**75.31**	83.34	87.23	39.82(+6.24%)
Ours	**17.80**	**17.91**	**18.15**	**33.36**	**35.51**	**40.28**	72.58	**83.77**	**89.09**	**45.16(+20.49%)**

TABLE 6 | Complexity and efficiency analysis of PKDNet.

Method	Total params (M)	Trainable params (M)	FLOPs (G)	Training time (s)	Inference (ms/img)
EPT	85.81	1.94	15.40	56.18	3.65
PKDNet	173.16	1.45	18.76	62.37	4.72

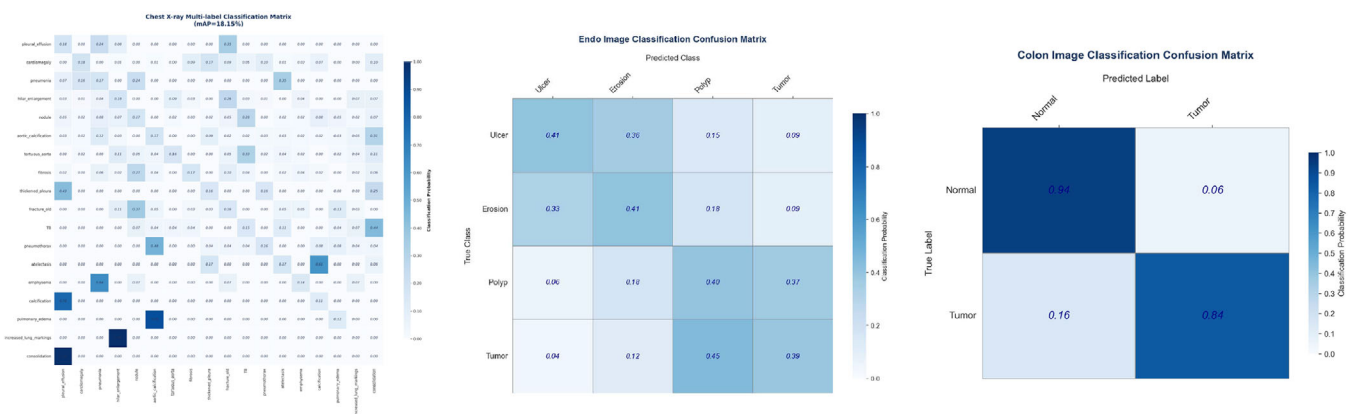


FIGURE 6 | Confusion matrices of PKDNet on different medical datasets. Class-level prediction results on the ChestDR, Endo, and Colon datasets under representative few-shot settings. The reduced off-diagonal responses indicate improved discrimination between visually similar disease categories, demonstrating the effectiveness of semantic–structural fusion.

is well justified. These results demonstrate that PKDNet achieves an effective balance between model capacity and efficiency, making it suitable for practical deployment in resource-constrained medical scenarios.

4.8 | Qualitative Analysis

4.8.1 | Confusion Matrix

To further explore the classification behavior underlying the quantitative results, we analyzed the confusion matrices of different methods on the ChestDR, Endo and Colon datasets. Figure 6 provides a class-level visualization of prediction errors under representative few-shot settings, enabling a fine-grained assessment of inter-class confusion patterns. The Endo dataset is especially difficult because different types of lesions look very similar. From the confusion matrix, it is clear that baseline methods exhibit distinct confusion between *ulcer* and *erosion*, as well as between *polyp* and *tumor*. These mistakes show that with only a few examples, it is hard for models to tell apart fine differences in texture and shape. In comparison, PKDNet makes far fewer errors between these similar classes. This improvement is clear from the stronger diagonal values and weaker off-diagonal values in its confusion matrix. This improvement suggests that the joint modeling of semantic cues (lesion type) and structural details (local texture and boundary characteristics) enhances fine-grained discrimination in endoscopic images. For the Colon dataset, which involves binary classification between normal tissue and tumor regions, baseline methods primarily suffer

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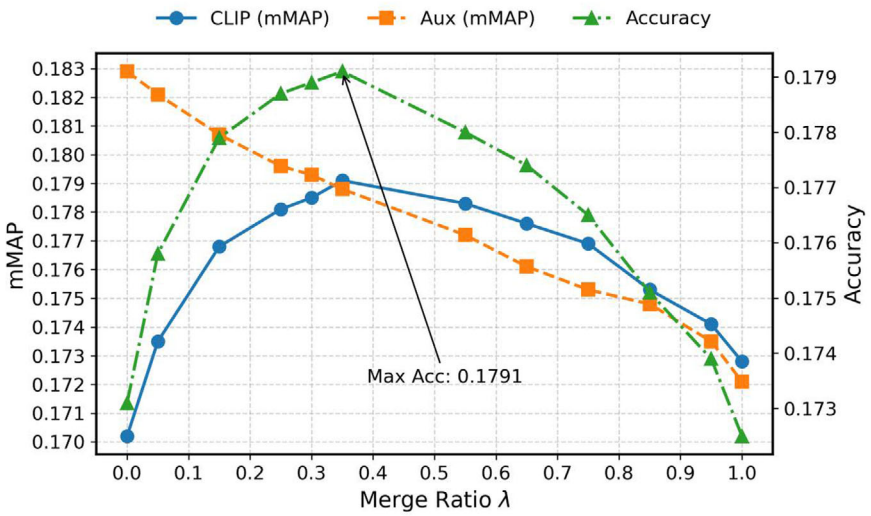


FIGURE 7 | Effect of fusion weight λ on model performance. Classification accuracy and mMAP of PKDNet under different values of the fusion weight λ, which balances the contributions of the CLIP-guided semantic branch and the enhanced ResNet branch during training. The model achieves optimal performance at an intermediate λ, indicating the necessity of semantic–structural complementarity.

from false negative errors, where tumor samples are incorrectly predicted as normal. The confusion matrix of PKDNet shows a noticeable reduction in such false negatives, accompanied by a higher true positive rate for tumor detection. This behavior aligns with the quantitative gains observed in the 5-shot and 10-shot settings, indicating that PKDNet improves sensitivity to pathological structures without introducing excessive false positives. The ChestDR dataset poses additional challenges due to its multi-label nature and severe class imbalance. The confusion matrix reveals that baseline methods frequently confuse diseases with overlapping radiographic appearances, such as *fibrosis*, *thickened pleura*, and *pleural effusion*, leading to elevated false negative rates for minority classes. PKDNet exhibits a more balanced error distribution, with reduced confusion among these disease categories and improved recognition of underrepresented classes. This indicates that semantic–structural fusion facilitates the disentanglement of subtle disease-specific patterns in chest X-ray images, even when label supervision is extremely limited.

Overall, the confusion matrix analysis shows that PKDNet reduces clinically meaningful inter-class confusion across modalities, complementing the quantitative results.

4.8.2 | Effect of Fusion Weight

We further investigate the influence of the fusion weight λ on the performance of PKDNet, where λ controls the relative contribution between the CLIP-guided semantic branch and the auxiliary enhanced ResNet branch during training. Figure 7 reports the variation of classification accuracy and mMAP with respect to $\lambda \in [0, 1]$. When λ approaches zero, the optimization is dominated by the CLIP-based semantic supervision. In this setting, the model performs poorly, which indicates that while CLIP can provide useful higher-level semantic information, it does not suffice for medical imaging because it cannot represent all finer pathological features of classifying medical images simply on semantically aligned images alone. Because of this, the auxiliary branch exerts virtually no influence, and the model's maximum level of class separation is relatively low. When the value of λ increases from 0 to around 0.35, both accuracy and mMAP steadily improve. This improvement indicates that structural information from the enhanced ResNet branch is gradually incorporated into the model. These structural features effectively complement the global semantic representations provided by CLIP. The best performance appears at $\lambda = 0.35$, where the model reaches its highest accuracy of 0.1791. At this point, semantic guidance and structural information are well balanced, and neither branch overwhelms the learning process.

However, when λ exceeds 0.35, model performance begins to decline. In this range, training places too much emphasis on low-level structural features, which reduces the influence of CLIP's semantic guidance. Consequently, the model becomes more sensitive to small visual differences and less stable when dealing with ambiguous semantic information. This change is reflected in the downward trend of both accuracy and mMAP.

Overall, the results indicate that an intermediate fusion weight yields the best balance between semantic guidance and structural discrimination.

4.8.3 | Sensitivity Analysis of Fusion Parameters

We further evaluate the sensitivity of the key fusion hyperparameters α and β. As shown in Table 7, the proposed method achieves the best performance when $\alpha = 0.5$ and $\beta = 0.4$. The results also show that the model remains relatively stable within a moderate range of values, indicating that the proposed fusion design is robust and reproducible. Specifically, when α is too small, the structural branch contributes insufficiently to the final prediction. In contrast, overly large α values overemphasize local structural cues and slightly degrade performance. Similarly, moderate values of β provide the most effective uncertainty

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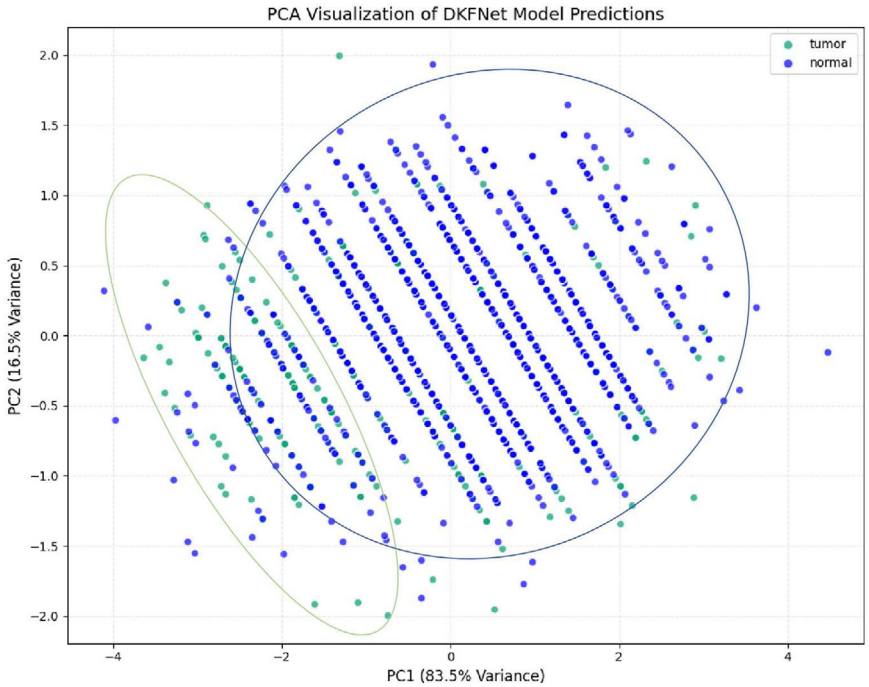


FIGURE 8 | This image shows the PCA visualization of the Colon dataset.

TABLE 7 | Sensitivity analysis of the fusion hyper parameters α and β under the 5-shot setting. The best performance is achieved at $\alpha = 0.5$ and $\beta = 0.4$.

α	Test mAP	β	Test mAP
0.10	0.1721	0.10	0.1748
0.25	0.1764	0.20	0.1773
0.50	**0.1791**	0.40	**0.1791**
0.75	0.1782	0.60	0.1786
1.00	0.1756	0.80	0.1769
		1.00	0.1745

modulation, while extreme values either suppress or exaggerate the structural contribution.

4.8.4 | Feature Distribution Analysis

To examine the differences in feature representations between classes, a principal components analysis (PCA) visualisation of model output is generated. In the PCA visualisations presented in Figure 8 the first two component axes represent 83.5% and 16.5% of the total variance explained, respectively.

The principal component (PC) visualisations show that, within the PCA space, the examples of each class group together into distinct clusters; while the normal examples are grouped closely together in a compact cluster, the tumour example values occupy far more of the PCA space than the normal examples and have wider distributions (more variation between examples). This disparity in distribution is consistent with clinical observations that tumours have many different types of visual appearance.

The PC visualisations also show that the two classes possess a small amount of overlap (aligned to different PC axes), which appears mainly at the boundaries of each class. This suggests that the feature representation of each class captures useful information to enable a correct classification of normal tissue and pathological tissue.

As the PC visualisations show, the more dispersed nature of the values for each tumour example, particularly along PC1, suggests that this component most likely encapsulates key semantic information related to the existence of disease. In contrast, the nature of the values in PC2 captures specific structural dissimilarity within each class.

4.8.5 | Failure Case Analysis

To further investigate the limitations of the proposed PKDNet, we present a representative failure case in Figure 9.

This example is particularly challenging due to the presence of multiple co-existing thoracic abnormalities, making it a complex multi-label scenario. In addition, the image exhibits significant structural distortions caused by medical devices, as well as low contrast and blurred boundaries in the lung regions.

Under such conditions, the structural branch struggles to extract reliable local features due to the interference of non-anatomical patterns, while the semantic representations derived from CLIP become less discriminative when multiple disease patterns overlap.

As a result, the fusion mechanism is unable to effectively reconcile conflicting cues from the two streams, leading to incorrect predictions.

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FIGURE 9 | Representative failure case of PKDNet on a multi-label chest X-ray image. The image contains multiple co-existing thoracic abnormalities and severe structural interference caused by medical devices, which increases semantic ambiguity and degrades the reliability of local structural cues.

This case highlights a key limitation of the current framework: its performance may degrade when both semantic and structural signals are simultaneously unreliable or heavily corrupted.

5 | Discussion

The experimental results show that PKDNet achieves different levels of improvement across modalities, highlighting both the advantages and the limits of semantic–structural fusion in cross-domain few-shot medical image classification.

The Endo dataset contains a large volume of highly detailed textures, and its images accordingly exhibit many local high-frequency color variations. Because these visual characteristics match well with the enhanced ResNet auxiliary branch, the ResNet's auxiliary branch is effective at capturing detailed fine structure while also benefiting from additional semantic information provided by CLIP. The strong match between the visual characteristics of the dataset images and the features extracted from the ResNet's auxiliary branch is the primary reason for the very large performance improvements observed by PKDNet in the Endo dataset setting.

On the ChestDR dataset, PKDNet consistently outperforms baseline methods, but with relatively small improvements. Chest X-ray images are mainly grayscale and depend more on global anatomical structures than on local textures. In this case, CLIP's semantic features play a major role, while the auxiliary branch adds only limited extra discrimination. As a result, the advantage of semantic–structural fusion decreases, especially as more labeled samples become available.

While PKDNet achieves competitive results on Colon data, it does not always outperform existing methods in very low-shot settings. However, significant variations in viewpoint, lighting, and deformations in colonoscopy images, with very limited supervision, make uncertainty estimation for feature fusion less reliable, thereby reducing the contribution of auxiliary structural features.

Overall, these results illustrate that PKDNet is most effective when semantic information and structural features complement each other. When either global semantics or local structures dominate, the benefit of fusion becomes limited. Future work may investigate geometry-aware feature extraction, uncertainty-based gating strategies, and domain-specific pretraining to further improve robustness across different medical imaging modalities.

6 | Conclusions

This paper proposes PKDNet, a dual-modal knowledge fusion framework for FSL. By integrating CLIP-based prompt learning with an enhanced ResNet backbone through uncertainty-aware fusion, PKDNet effectively combines global semantic representations with fine-grained structural features to address semantic ambiguity and data scarcity. Extensive experiments on multiple medical datasets demonstrate that PKDNet consistently outperforms existing adaptation and prompt-based methods while maintaining parameter efficiency and training stability. These results highlight the importance of explicitly modeling semantic–structural complementarity when adapting foundation models to diverse medical imaging modalities under limited supervision.

Author Contributions

Yingying Huang: funding acquisition, writing – review and editing. **Zihan Shao:** methodology, formal analysis, visualization, writing – original draft. **Renfang Wang:** investigation. **Xiaozhe Gu:** data curation. **Jiashun Wu:** validation. **Hong Qiu:** supervision.

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Conflicts of Interest

The authors declare no potential conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in references [40, 41]. The code will be made publicly available after the work is accepted.

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